LESSON 6.7

Find Factors and Multiples

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the greatest common factor and the least common multiple of two numbers, and use each to solve a real problem.

WHY THIS MATTERS The station quartermaster has two jobs today. First, split the pencil and notebook stock into identical combo packs with nothing left over — that is a factor question. Second, work out when two supply runs that repeat on different cycles will land in the same week — that is a multiple question. Same lesson, opposite tools. Today the stock is 24 pencils and 36 notebooks. One supply run arrives every 6 days and another arrives every 8 days.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Factor *Spanish: Factor*	A whole number that divides another number evenly, with no remainder.	1, 2, 4, 8, 16 divide 16.
Multiple *Spanish: Múltiplo*	The result of multiplying a number by 1, 2, 3, and so on — where you land when you skip-count by it.	Multiples of 8: 8, 16, 24, 32, 40 …
Greatest common factor (GCF) *Spanish: Máximo común divisor (MCD)*	The largest number that is a factor of two or more numbers. It answers “how many equal groups can I make?”	48 and 36 share 1, 2, 3, 4, 6, 12 → GCF = 12
Least common multiple (LCM) *Spanish: Mínimo común múltiplo (mcm)*	The smallest number that is a multiple of two or more numbers. It answers “when will these line up again?”	6, 12, 18, 24 and 8, 16, 24 → LCM = 24
Factor pair *Spanish: Par de factores*	Two factors that multiply to make a number. In a word problem, one is the number of groups and the other is the size of each group.	2 × 12 = 24
Prime factorization *Spanish: Descomposición en factores primos*	Writing a number as a product of prime numbers, usually with a factor tree. Multiplying the primes two numbers share also gives the GCF.	24 = 2 × 2 × 2 × 3

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find the GCF of 20 and 30.

1. Factors of 20: 1, 2, 4, 5, 10, 20.
2. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30.
3. Shared factors: 1, 2, 5, 10. The greatest is 10.

Answer: GCF = 10

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A factor tree breaks each number into primes. 48 = 2 × 2 × 2 × 2 × 3, and 36 = 2 × 2 × 3 × 3.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The factors of _____ are _____ .
- The factors that both numbers share are _____ .
- The greatest common factor is _____ .
- I used the GCF here because the problem asks for _____ .

Word bank factor | factor pair | common factor | greatest common factor | GCF | prime | prime factorization | divides evenly | list | share | greatest | equal groups

WATCH OUT!

A common mistake

A common mistake is reaching for the wrong tool: listing FACTORS for a problem about repeating cycles. Ayasha's classes repeat every 6 and 8 weeks, so the weeks they land on are multiples

LESSON 6.7

Find Factors and Multiples

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each number by its relationship to 10.

Groups: **A factor of 10 only** | **A multiple of 10 only** | **Both a factor and a multiple of 10**

_____ 2

_____ 20

_____ 5

_____ 30

_____ 10

_____ 50

HINT Check both relationships: does this number divide 10 evenly, and does 10 divide this number evenly?

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which list shows all the factors of 12?

(A) 1, 2, 3, 4, 6, 12

(B) 1, 12

(C) 2, 3, 4, 6

(D) 12, 24, 36, 48

HINT Ask: what divides 12 with nothing left over?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which list shows the first four multiples of 8?

(A) 1, 2, 4, 8

(B) 2, 4, 6, 8

(C) 8, 16, 24, 32

(D) 8, 9, 10, 11

HINT Skip-count by 8 out loud.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the greatest common factor of 12 and 18?

(A) 2

(B) 3

(C) 6

(D) 36

HINT List the factors of each number.

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

What is the least common multiple of 4 and 6?

(A) 2
(B) 10
(C) 12
(D) 24

HINT List multiples of each number.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Find the error

1. **Problem** Find the GCF of 12 and 18.
2. **Step 1** Factors of 12: 1, 2, 3, 4, 6, 12
3. **Step 2** Factors of 18: 1, 2, 3, 6, 9, 18
4. **Step 3** The GCF is 36, because 36 is in both times tables.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Compare Step 3's answer to the two numbers themselves.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Both problems say "common." How do you decide whether a problem wants the GCF or the LCM?

SENTENCE STARTER If the problem ___ something into equal groups, I use the ___. If it asks when things ___ again, I use the ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.7

Find Factors and Multiples

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

The store has 48 pencils and 36 notebooks. Complete the factor lists, then find the GCF.

Number	Factors	GCF
48	1, 2, 3, 4, 6, 8, 12, 16, 24, 48	
36		

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each real problem by the tool that solves it.

Groups: Use the GCF | Use the LCM

_____ Split 48 pencils and 36 notebooks into identical packs

_____ Two classes repeat every 6 and 8 weeks — when do they meet?

_____ Cut two ribbons into the longest equal pieces with none left over

_____ Hot dogs in packs of 10, buns in packs of 8 — when do they match?

_____ The answer is at most the smaller number

_____ The answer is at least the larger number

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Using prime factors, 48 = 2 × 2 × 2 × 2 × 3 and 36 = 2 × 2 × 3 × 3. What is their GCF?

(A) 2, the only prime they share

(B) 6, from one 2 and one 3

(C) 12, from the shared 2 × 2 × 3

(D) 144, from multiplying all the primes

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Ayasha has painting every 8 weeks and pottery every 6 weeks, both this week. When do they next fall in the same week?

(A) In 2 weeks

(B) In 14 weeks

(C) In 24 weeks

(D) In 48 weeks

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Problem** Ayasha has painting every 8 weeks and pottery every 6 weeks. When do both fall in the same week?
2. **Step 1** Factors of 8: 1, 2, 4, 8
3. **Step 2** Factors of 6: 1, 2, 3, 6
4. **Step 3** The greatest common factor is 2, so both classes meet in week 2.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Both problems say “common.” How do you decide whether a problem wants the GCF or the LCM?

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LESSON 6.7

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find both the GCF and the LCM of each pair.

Numbers	GCF	LCM
6 and 8	2	24
12 and 18		
4 and 10		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the error

1. **Problem** Find the LCM of 6 and 8.
2. **Step 1** Multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48
3. **Step 2** Multiples of 8: 8, 16, 24, 32, 40, 48
4. **Step 3** Both lists contain 48, so the LCM is 48.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Reginald is having hot dogs for 24 guests and wants the same number of hot dogs and buns, with none left over. Hot dogs come in packs of 10 and buns in packs of 8. How many packs of each should he buy, and how many will he have? Explain which tool you used and why.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Both problems say "common." How do you decide whether a problem wants the GCF or the LCM?

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